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e-Supply network coordination: the design of intelligent agents for buyer-supplier dynamic negotiations

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Abstract This paper concentrates on the problem of automated negotiations in e-supply network coordination (e-SNC). Supply network is considered as a collection of agent-mediated decisions and coordination mechanisms in a web-based environment. The proposed agent based coordination model is composed of two negotiator agents and a coordinator agent. The coordination mechanism begins with determining the feasible and most promising partners in the network based on the similarity of profiles. In response to autonomy levels and conflict objectives, the model allows agents to negotiate in a cooperative manner through an iterative process of generating offers for re-establishing global optimality. The dynamic negotiation model is then defined based on the protocol, rule of bargain, proposal generation, and dynamic strategy. To illustrate the model efficiency, a prototype system has been modeled and is compared to the normal tendering mechanism. The validation results confirm the model efficiency in providing coherent decision making in a dynamic environment.

Keywords e-Supply network coordination · Feasibility analyzer · Automated negotiations

Introduction

Complex economic activities often involve interrelated and multiple exchange relationships among the parties involved in the respective business. In reality, products are no longer produced and consumed within the same geographical area. Different parts of a product may, and often do, come from all over the world. This creates longer and more complex supply chains and thereby it changes the requirements within supply chains. Hence, the traditional linear supply chain is converted into a supply network in which collaboration between manufacturers and their supply chain partners will be improved (Lu and Wang 2008).

Recent technological innovation provides us with additional opportunity to deal with such a complex network. For instance, information technology provides an infrastructure for integrating the internal and external activities of a company, so it connects the geographically distant supply chain members together to form a network system. On the other hand, electronic market (EM) has created opportunities for a meaningful investigation of the business relationships in supply networks (Mahdavi et al. 2008). Hence, e-Supply Network (e-SN) can be defined as a new structure which is enhanced by the trend of market globalization, information technology, and automation of different echelons' behavior (Mohebbi and Shafaei 2009). It emphasizes on the cooperation among supply echelons, and realizes the dynamic nature of information, capital, and materials flows with the assistance of electronic commerce and intermediary services. One key feature of the e-SN is the dynamic alteration of interaction structure, and the nature of network attributes/criteria as represented by informational flow. New buyers and sellers join and existing buyers and sellers move out, new products are introduced or replace existing one, and product evaluation attributes keep change over time (Mahdavi et al. 2009).

In addition to the dynamic nature of an e-SN, there is a couple of key factors affecting the performance of an e-SN. This includes coordination among the network members and negotiation among those partners whose interrelations have been established but need to be enhanced.

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SNC can be defined as the coordination of distributed decision making of organizations or participants on material flow, information flow and financial flow. Regarding the SNC, some existing approaches have focused on the design of coordination mechanisms for a two-echelon supply chain: buyers on the one side and suppliers on the other side. A buyer and a supplier together constitute a simple two stage supply chain. The idea of joint optimization for buyer and vendor was initiated by Goyal (1977) and later reinforced by Bannerjee (1986). Afterwards, Weng (1999) demonstrated that when both parties are coordinated, the order quantity and joint profit will increase and hence the selling price will decrease. Aviv (2001) also investigated the effect of collaborative planning on supply chain performance. The coordination of a supply chain, however, requires accurate and timely information about the operational decisions and activities to be shared among all members to deal with uncertainties (Li and Wang 2007). Such information is those which the parties are willing to share with their partners and hence does not include the member's confidential information. Schneeweiss and Zimmer (2004) presented an operational coordination mechanism between a producer and a supplier within a supply chain having private local information. This investigation finally gives recommendations for the design of a supply contract. Chu and Leon (2009) also investigated the problem of coordinating assembly inventory systems with private information system so that the objective function and cost parameters of each facility in two-echelon supplier-buyer model are regarded as private information. It can be, therefore, concluded that the main issue on SNC is to establish a suitable structure of information which should be shared among the network members. In other word, assuming a complete information flow without adequately capturing the dynamics and uncertainties of the environments may lead to an over simplified and restricted models in SN.

As addressed before, the second factor affecting the performance of a SN is negotiation among the partners. Negotiation has long been recognized as a time-consuming process since all parties involved intend to pursue their own interests in the face of conflicting goals. Furthermore, even in the most simple of negotiations, individuals frequently reach sub-optimal agreements (Raiffa 1982). Indeed, in a web-based environment, the ability to rapidly identify suitable partners and effectively coordinate them through the net is crucial to the success of SN. Hence, although an e-marketplace eliminates the geographic obstacles to a certain extent, the barriers of culture, ego, and pride associated with human-based negotiations (Walton and McKersie 1965) can still constraint the efficiency of the marketplace. In addition, the sheer number of participants in the e-marketplace introduces difficulties when attempting to find, and negotiate with potential buyers or suppliers (Cheng et al. 2006).

The implementation of automated negotiations conducted by software agents in the e-marketplace may overcome the difficulties in human negotiations. In fact, agents are offshoots of distributed artificial intelligence having a long-term goal of developing mechanisms and methods to enable agents to interact like human beings, or even better (Weiss 1999). It offers a new perspective of autonomous activity, interactivity, reactivity and pro-activity in an attempt to extend beyond speeding up the communications, calculations, and routine computation in business interactions (Wang and Wang 2006; Wang et al. 2005). Accordingly, effective coordination and negotiations play an important role in enhancing an e-SN performance. However, the review of the researches in SN reveals that quantitative investigations in modeling both coordination mechanism and negotiation process in SN have not deeply been addressed.

In this study, coordination mechanism for information and material flows and negotiation process among buyers and suppliers in a dynamic SN has been investigated. We propose an agent-based mathematical model in a web-based environment. Each buyer and supplier in the SN possesses different attributes such as desired delivery time, price and etc. Therefore, in order to achieve a global optimal solution, each party should consider various trade-offs for decision making. Building rich and accurate profiles of buyers and suppliers based on their priorities is of crucial importance in e-SNC. The proposed coordination mechanism among buyers and suppliers begins with finding most promising partners based on the similarity of users' profiles in a multidimensional space defined by negotiable attributes. This leads to determine the feasible partners in the network. Afterwards, the negotiation process among the feasible partners is formulated so that the material flow and conflicting goals of the members are taken into account. This approach allocates the benefits of the mechanism to all partners and optimizes the network global objective function as well. In the proposed model, corresponding agents possess private information sets and process data on-line, while the aggregate value and cost of the whole network is to be managed real-time using the overall model. The agents' policies for negotiation are created by corresponding rules in conjunction with network overall objective functions. This approach allows us to take the advantage of the developments and diffusion of modern information and network technologies.

The rest of this paper is organized as follows: In Sect. "Background", the literature on intelligent agents, and related approaches for automated negotiations in SNC are reviewed. In Sect. "Proposed e-SN structure", e-SN architecture, and detailed mechanism for the model are presented. Section "Validation and Performance Evaluation" describes the implementation details for simulation. Next, a statistical analysis is conducted to validate the proposed approach.

Finally, the conclusions and some guidelines for further research are given in Sect. "Conclusions".

Background

Intelligent agents in SNC

Both the functionality and structure of an e-SN may vary from highly to loosely integrated levels. The main interest of managers is to ensure that the overall cost is reduced and operations among various systems are integrated through coordination (Fazel Zarandi et al. 2008). Hence, decision making and coordination in such network is usually complex and frequently partitioned into sub-problems. It requires that companies be able to effectively and efficiently coordinate activities such as production and transportation across supply chains that are dynamically set up in response to constantly changing and increasingly customized market requirements (Sadeh et al. 2003).

Lately, Internet-based technologies such as web services have been emerging. E-business has been a powerful and compelling enabler of supply chain integration across a wide range of industries (Kaplan and Sawhney 2000). However, despite the merits of these technologies, there exist some limitations in flexibility and dynamic coordination of distributed participants in supply chains. The multi-agent systems (MASs) are alternative technologies for automated decisions and coordination in SN because of the certain features such as distribution, collaboration, autonomy, and intelligence. An agent is a software entity, which is characterized with environment awareness, ongoing execution, autonomy, adaptiveness, intelligence, mobility, anthropomorphism and reproduces. Thereby, a MAS consists of a number of agents, which interact with one another in order to carry out tasks through cooperation, coordination and negotiation (Wooldridge 2002). They can monitor and retrieve useful information, and do transactions on behalf of their owners or analyze data in the global markets (Xue et al. 2007). Therefore, the major challenge of the agent-based supply coordination is to find a global solution for the composite service such that all agents find the solutions that satisfy not only their own constraints but also inter-agent constraints (Liu et al. 2002).

Supply chain scholars have taken various complementary perspectives when investigating coordination and information sharing within a supply chain, including MASs. Kimbrough et al. (2002) proved that a team of agents could eliminate the bullwhip effect through sharing information such as order quantity. Ito and Abadi (2002) applied agent technology to warehouse systems in order to enhance supply chain success. Sadeh et al. (2003) introduced MASCOT, Multi-Agent Supply Chain Coordination Tool, an architecture that aims at providing a framework for dynamic selection of supply chain partners and for coordinated development and manipulation of planning and scheduling solutions at multiple levels of abstraction across the supply chain. In their proposed approach, MASCOT places a particular emphasis on mixed-initiative problem solving, enabling users to flexibly select among a broad range of interaction regimes. Lu and Wang (2008) introduced the network economy as a new economic system and built an equivalent topological structure of it. Hence, the relationship between input and output elements have been introduced based on this topological structure. According to these relationships, they studied a framework of supply chain in network economy through MASs. Wang et al. (2008) proposed an agent-mediated approach to on-demand e-business supply chain integration. Each agent works as a service broker, exploring individual service decisions as well as interacting with each other for achieving compatibility and coherence among the decisions of all services. Mahdavi et al. (2009) proposed a dynamic model for the agent-based SN as a solution for coordinating buyers and sellers. They utilized the concept of users' profile in the network and presented an optimization model in conjunction with what-if simulation module to obtain mutually compatible solutions. In this model, however, the negotiation among the buyers and suppliers to enhance their relationship has not been discussed.

In our work, e-SNC is mapped as an agent-mediated coordination and negotiation problem in a web-based environment. In the next section, we discuss the use of automated negotiations in SNC in detail.

Automated negotiations in e-SNC

Automated negotiation can be viewed as a search process (Choi et al. 2001; Oliver 1997) in which negotiator agents in SN search for a mutually acceptable partner to establish/re-establish global consistency and optimality (Mohebbi and Shafaei 2009). The basic assumption for the negotiations in SN is that each member of the network normally has its own policy which might be on the contrary to that of the others. Hence, a coordination mechanism is required to be activated for aligning and synchronizing the contributions of the SN members to the whole business process.

In a typical SN, the supplier and the buyer negotiate on price and lot size, and the outcome depends on the balance of power between the two parties. SN scholars have also taken various perspectives when investigating automated negotiations. Faratin et al. (2002) proposed an automated negotiation mechanism based on the concept of issue (attribute) tradeoffs. Their algorithm performed an iterated hill-climbing search in a landscape of possible contracts. Karimi et al. (2007) proposed a new multi-attribute procurements auction for the agent-based supply chain as a solution for negotiation between buyers and sellers. They utilized the approach of game theory and simulation to obtain and compare the optimal strategy based on their risk attributes. Li and Yang (2008) proposed a general bilateral automated negotiation process based on the prioritized fuzzy constraint satisfaction problem as well as a negotiation strategy based on constraints relaxation and genetic algorithm. Lee and Yang (2009) presented a neural network approach for forecasting the suppliers' bid prices in supplier selection negotiation process. Nassiri-Mofakham et al. (2009) proposed a heuristic personality-based bilateral multi-issue bargaining model in electronic commerce which deals with bargaining of single buyer and single seller. Among the automated negotiation studies in the literature, Buttner (2006) proposed taxonomy for classifying the automated negotiations based on the type of process, theoretical foundation, structure, and restrictions. The results of the paper revealed the manifold addressing of the negotiation challenges, but mainly related to the structure and the process of the negotiation. Most of the studies considered price and lead time independently and as a two main negotiable attributes. However, the issue of material flow in SNC can complicate the decisions in a negotiation as it generally interacts with other issues such as price and lead time.

In a SN, besides the price attribute, the products are normally procured based on a number of other attributes such as product quality, payment terms, and delivery conditions. These attributes are also commonly treated as negotiable criteria. The details of these attributes are presented in Table 1. Categorization of these attributes based on the semantic definition is an important procedure within the knowledge learning loop for e-SNC. It helps users flexibly select attributes and/or attribute values for a particular analysis session. The importance levels, or weights, assigned to these criteria may vary between the two parties involved in the negotiation, and hence trade-offs can be made between different criteria such that both parties can reach a mutually beneficial agreement. On the other hand, the sheer volume of customer information and increasingly complex interactions of customers and suppliers has propelled the clustering of documents to the forefront of making the coordination effective.

The attributes of each category in SN may be identified based on similar patterns. To avoid evaluating on similar attributes repeatedly, ones should partition them into several clusters. It can help managers to extract effective attributes from buyers' perspectives for each product. Indeed, negotiations in SN only with promising opponents minimize the occurrence of pointless negotiations and increase the rate of successful contracts. Hence, decisions should be driven by only effective attributes which guarantee a speedy delivery of relevant contents.

Here, we extend the Mahdavi et al. (2009) work and propose an integrated model for effective coordination and negotiation among buyers and suppliers in e-SN. The users' profile in the network is designed so that to be applicable for enlarged network with sheer number of users and respective attributes. We also propose software agents including two negotiator agents and a coordinator agent that negotiate in a cooperative manner in attempt to enhance the efficiency of automated negotiations in dynamic situations. We first apply a fuzzy c-means clustering method to partition categories of attributes into several clusters and select the representative ones from these clusters. Then a feasibility analyzer is designed to study the possibility of interaction between agents. The coordinator agent matches buyers and suppliers by finding the most similar profiles to each participant. In this manner, each negotiator agent need only negotiate with the few opponents considered to be the most promising. In practice, depending on the participant's criteria for price and delivery, different rules may be selected by each participant. For this purpose, in the proposed negotiation process the function of price proposal is considered to be dependent on the lead time in order to generate suitable rules for negotiators while considering the total welfare of the network. Postulating the Buttner (2006) taxonomy for automated negotiation models, our proposed negotiation mechanism for e-SNC, is full-automated, goal-oriented, argumentation-based, and time-dependent, as well as its structure is double-sided multilateral negotiation, multi-attributes, mediated, and integrated (win-win approach).

Table 1 Negotiable factors in SN

Category	Attributes
Performance	Installation ease
	Life cycle
	Guarantee
Price	Development and tooling cost
	Production and packaging cost
	Transportation cost
Delivery and customer service	Order processing time
	Production and delivery lead time
	Tardiness penalty
	After sale service
Quality	Dimensional
	Mechanical characteristic
	Metallurgical characteristic
	Aesthetics
	Scrap rate
Manufacturing domain	Products diversity
	Products' upgrading
Stability and confident	Information confident
	Commercial confident
	Commitment to long-term relationship

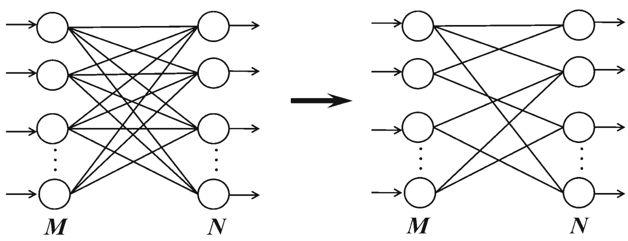

Fig. 1 Supply network structure

In the next section, we will illustrate how the proposed approach can help us to achieve rich and accurate profiles of customers and suppliers through the dynamic clustering of attributes.

Proposed e-SN structure

In the proposed approach, a SN has M input elements (i.e. buyers with corresponding attributes) which enters the network and the respective requirements will be handled by N output elements (i.e. suppliers) via the functions of network structure (Fig. 1). According to the input layers attributes and output layer capabilities in terms of the scope of work and available capacity, various possible relations can be established. Hence, for this enlarged network structure, there are N^M possible relationships among the members. The link between each couple of nodes may perform a specific function, such as investing, manufacturing, transporting, or purchasing in SN. On the other hand, SN is a complete open system and thereby it communicates with the environment over time. Hence, although utilizing the method of procurement, manufacturing, inventory and transportation control tools, inventories and order-to-cash cycles can be kept on a relatively low level (Lu and Wang 2008); the sheer volume of each layer information and increasingly complex interactions among the nodes has jeopardized the existing approaches for supply chain management. Therefore, in this study, the initial two-echelon SN is converted into enhanced SN so that all possible relations among buyers and suppliers are investigated and ultimately each node will negotiate only with most promising nodes (Fig. 1).

In addition, it is assumed that suppliers are located in different nations with a vast network of forward agents. Indeed, consider the family of products that a buyer would like to procure from an electronic market for which there are some pre-qualified suppliers available to supply. Usually a supply link has to be embedded into the entire SN. For instance, the supplier might not have just one but several other producers he/she is in contact with. In capturing this influence of the full network, the availability of the suppliers' capacity is considered to be stochastic.

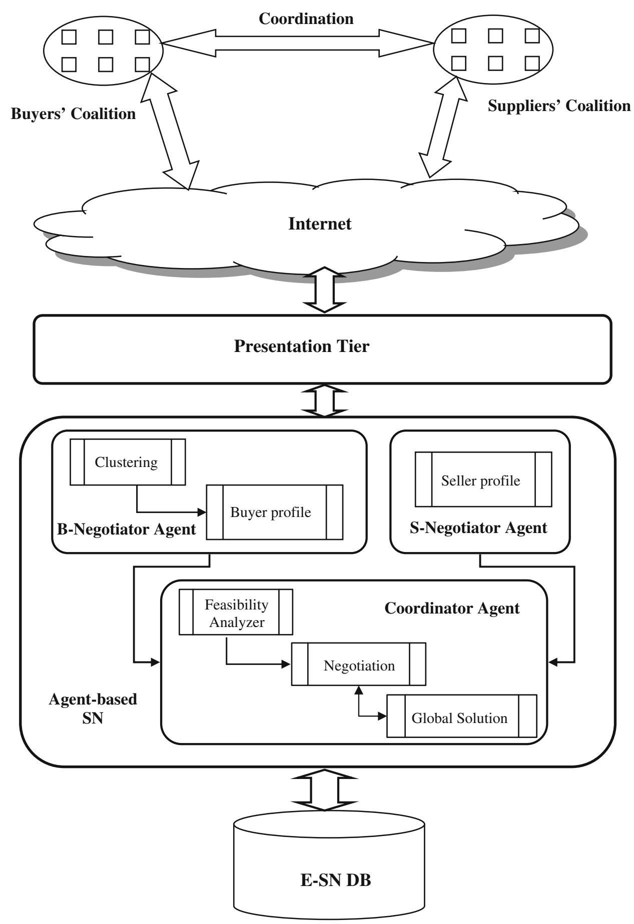

Fig. 2 Agent-based e-SN architecture

In the next section, we describe the functionality of multi-transaction agents which play the most important role in our proposed supply system. This helps to create a more coherent mechanism for resolving conflicts among functional units.

Multi-agent system architecture

The overall architecture of the proposed agent-based network is displayed in Fig. 2. One problem commonly faced by the agents acting in SNs is that of negotiating with customers in order to sell goods. Such negotiations are often handled through tender in which suppliers submit sealed bids in response to requests for quotes (RFQs) from customers. This situation becomes particularly difficult when sellers must bid in multiple tenders simultaneously, because an agent cannot await the outcome of one auction before bidding in another.

Here, each agent is autonomous in making decisions on behalf of each firm. That is, each agent autonomously processes and collects information, and makes decisions to maximize the profit of the corresponding firm. An additional agent, the *Coordinator Agent*, is introduced as a central coordinator to enhance the coordination amongst the software agents (Luo et al. 2001). Each agent possesses its own information that is not shared with other agents. Each agent is responsible for making and monitoring its decisions; it also can incorporate a change of goals caused by supply uncertainties. Next, we elaborate on the functionality and interactive behavior of agents involved in the decision making and coordination for e-SN.

B-Negotiator Agent It serves as a buyers' coalition and is responsible to utilize the costumer information including orders and individual objectives of buyers, quantitative and qualitative attributes related to customers' evaluation. The goal of the agent is to optimize the objective functions of buyers in the SN. Using clustering module, *B-Negotiator Agent* first extracts the effective and most desired attributes of buyers. Then, it preprocesses and builds the customer profiles.

S-Negotiator Agent It serves as a suppliers' coalition and receives buyers' attributes, offered prices of suppliers based on capacity and inventory carrying cost, quantitative and qualitative attributes, related to suppliers' perception. Then, it preprocesses and builds supplier profiles. The goal of an *S-Negotiator Agent* is to optimize the individual objectives of suppliers in order to satisfy the relevant demand and capacity.

Coordinator Agent It facilitates additional levels of coordination, negotiation and selection of business counterparts by receiving both buyers' and suppliers' information. The *Coordinator Agent's* main priority is to maximize the overall SN profits by coordinating with the other agents. Using the feasibility analyzer module, the agent determines the most promising counterparts. Afterwards, the feasible partners participate in the negotiation process so that the goodness of the proposals will be evaluated by the global solution module. In the next section, we will illustrate how to acquire rich and accurate profiles in an e-SN.

Analytical modeling

With the view of the above addressed points, the notations of variables and concepts to be used in the rest of the paper are as follows:

- m: number of buyers in the network
- k: number of suppliers in the network
- g: number of products which is to be transacted
- n: number of key attributes
- h: number of category keywords
- $\tilde{c}_i^{dr}$: the fuzzy membership value of ith buyer corresponding to key attribute d in category r
- $\tilde{Y}_{ij}^{d}$: the fuzzy membership value of ith buyer for j^{th} supplier corresponding to dth key attribute
- $\tilde{a}_{ij}$: the fuzzy membership value of final priority level for ith buyer with respect to jth supplier
- $\tilde{\beta}_{ij}$: the fuzzy membership value of final priority level for jth supplier with respect to ith buyer
- CF_i^b: objective functions of ith buyer
- CF_j^s: objective functions of jth supplier
- $SNDM = [dsn_{ij}]$: the binary decision matrix

B-Negotiator Agent

Collect information

The *B-Agent* first designs a class of part families of the net as follows:

$$A = (A_1, A_2, \ldots, A_h) \tag{1}$$

The *B-Agent* then creates the array of bids, where a bid is defined as follows:

$$Bid_{iu} = [B\_id_{iu}, S\_t_{iu}, e\_t_{iu}, Loc_{iu}, D_{iu}] \quad \begin{array}{l} \forall i = 1, \ldots, m \\ \forall u = 1, \ldots, g \end{array} \tag{2}$$

Bid_{iu}, represents the bid of ith customer for uth product which contains of five elements B_id_{iu}: denotes the ID number of the customer which is associated with uth product; S_t_{iu} and e_t_{iu} denote the entry time and desired lead time of ith customer for uth product, respectively; Loc_{iu} denotes the desired destination of the ith buyer for uth product; D_{iu} is the demand of ith buyer for uth product.

Many attributes are involved in the process of supply chain coordination. Attributes evaluation is a process whereby the value of the attributes are evaluated, based on the preferences of the participants in the SN. All attributes are classified into two categories: quantitative and qualitative. Given an initial categories of attributes, the *B-Agent* receives all necessary information about the quantitative and qualitative attributes related to each product and seller from m customers electronically. The agent decides on n key attributes for all aspects of customers' perceptions. Then, vectors of key-attributes category are created as $CK^r = \{k_1, k_2, \ldots, k_{n_r}\}$. For each category, there are n_r attributes at time T_0 in the network. The eligible buyers may suggest or request some additional attributes for each product. Hence, a node can be designed in the network in order to receive these additional attributes and convey to the suppliers. The suppliers then check the practical justification and the frequency of use for the suggested attributes and verify them to add in the network. Each supplier should quantify these attributes for each product on behalf of the firm capacity and constraints. This information would then be placed in their catalogue on the network. The agent then designs a buyer-key attributes incidence matrix as $BKIM^r = \left[\tilde{c}_i^{dr}\right]$, where $\tilde{c}_i^{dr}$ represents the fuzzy value of ith buyer corresponding to the key attribute d in category r using linguistic terms. This value indicates the priority and evaluation of the buyer on a special aspect of a product. These linguistic terms are also divided into very low (VL), low (L), medium (M), high (H) and very high (VH). Assume that

Table 2 The importance degree

Very low	(0,0,20)
Low	(0,20.40)
Medium	(30,50,70)
High	(60,80,100)
Very high	(80,100,100)

all linguistic terms can be presented with triangular fuzzy numbers as shown in Table 2.

Some of these attributes are identified on similar patterns. To avoid evaluating on similar attributes repeatedly as well as reducing the computation time, key attributes should be partitioned into several clusters, and a representative indicator is selected from one cluster to be as an evaluation criterion. Based on this idea, we have applied the following process in the first phase of negotiation.

Finding the representative indicators from buyer-key attributes

Building rich and accurate profiles of customers based on their priorities is of crucial importance in e-SN. Customer profiles contain factual information about customers which can be used to describe their behavior. One problem is that many discovered rules are irrelevant or trivial. One should find a way to select the best set of priorities for each customer. For this reason, we suggest the following steps:

Step 0 Defuzzify each element of $BKIM^r$ matrix using centroid method.

Step 1 Normalize the elements of above matrix as follows: If c_i^{dr} belongs to benefit items, then

$$nc_i^{dr} = \frac{c_i^{dr}}{\sqrt{\sum_{i=1}^{m} (c_i^{dr})^2}} \quad \forall d = 1, \ldots, n_r \quad \forall r = 1, \ldots, h \tag{3}$$

If c_i^{dr} belongs to cost items, then

$$nc_i^{dr} = \frac{1/c_i^{dr}}{\sqrt{\sum_{i=1}^{m} (1/c_i^{dr})^2}} \quad \forall d = 1, \ldots, n_r \quad \forall r = 1, \ldots, h \tag{4}$$

In the above equations, nc_i^{dr} is the normalized preference value of ith buyer corresponding to the key attribute d in category r.

Step 2 Apply Fuzzy C-Means (FCM) to cluster the attributes of each category.

Step 3 Determine the optimal number of clusters using the criteria presented by Fukuyama and Sugeno (1989) as given in Appendix.

Step 4 Find the representative of each cluster so that the attribute with maximum membership value would be selected.

Step 5 Constitute the modified buyer-key attribute incidence matrix as $MBKIM = \tilde{c}_i^{\prime dr}$, where $\tilde{c}_i^{\prime dr}$ represents the preference fuzzy value of ith customer corresponding to the cluster representative d in category r.

Attributes have limited life. If a specific attribute was not used over certain period, then it is disqualified to remain in the network. The clustering approach is updated whenever changes appear in the structure of attributes.

Buyers' profile

After processing the first phase of negotiation, on the basis of a specified threshold value for each key term in corresponding category, the *MBKIM* matrix could be converted to a binary matrix $BMBKIM = \left[bc_i^{dr}\right]$, where $bc_i^{dr} = 1$ if $\tilde{c}_i^{\prime dr} > \alpha_d$ and $bc_i^{dr} = 0$ otherwise. α_d is the threshold value of the keyword d. Hence, we can obtain the weight of each attribute using the following formula.

$$w_d^b = \frac{\sum_i bc_i^{dr}}{\sum_i \sum_d bc_i^{dr}} \quad \forall d = 1, \ldots, n_r \tag{5}$$

Then, the *B-Agent* designs a buyer-supplier incidence matrix as $BSIM^d = \tilde{Y}_{ij}^d$, where $\tilde{Y}_{ij}^d$ represents the preference fuzzy value of ith buyer for jth supplier ($i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, k$) corresponding to the dth representative attribute.

To obtain the buyer's profile matrix, we can calculate the average weight of $BSIM^d$ matrix as below:

$$MBSIM = \begin{bmatrix} \tilde{a}_{11} & \cdot & \cdot & \tilde{a}_{1k} \\ \tilde{a}_{21} & \cdot & \cdot & \tilde{a}_{2k} \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \tilde{a}_{m1} & \cdot & \cdot & \tilde{a}_{mk} \end{bmatrix} \tag{6}$$

where, $\tilde{a}_{ij} = \sum_{d=1}^{n} \tilde{Y}_{ij}^d w_d^b$.

We observe that the value calculated for $\tilde{a}_{ij}$ as above represents the final priority level of ith buyer on jth supplier.

The agent exists in a dynamic environment and thereby, beyond the environmental attributes, it has a set of objective functions.

Let $CF_i^b = \{f_1^b, f_2^b, \ldots, f_l^b\}$ be the sets of conflicting objective functions of ith buyer. The agent receives the type of objective functions. That is, each agent autonomously processes and collects information, and makes decisions to optimize the objectives of the corresponding firm.

S-Negotiator Agent

Collect information

The *S-Agent* first creates the array of service, where a service is defined as follows:

$$Serv_{ju} = \left[S\_id_{ju}, S\_t_{ju}, e\_t_{ju}, Loc_{ju}, S_{ju}\right] \quad \begin{array}{l}\forall j = 1, \ldots, k \\ \forall u = 1, \ldots, g\end{array} \tag{7}$$

$Serv_{ju}$, the service of jth seller for uth product which contains of five elements: S_id_{ju} denotes the ID number of the seller which is associated with uth product; S_t_{ju} and e_t_{ju}denote the entry time and due date of jth seller for uth product, respectively; Loc_{ju} denotes the location of the jth seller for uth product; S_{ju} is the supply of jth seller for uth product.

Suppliers' profile

The *B-Agent* forwards all the information related to each product and representative attributes in the buyers' profile to all suppliers electronically. Then, agent obtains the fuzzy value of each supplier on the buyer keywords. By obtaining all information from supplier side, it creates the supplier-keyword incidence matrix. The agent designs a supplier-keyword incidence matrix as $SKIM = \left[\tilde{s}_j^d\right]$, where $\tilde{s}_j^d$ represents the fuzzy value of the jth supplier on the corresponding attribute. This value indicates the priority of supplier to satisfy that specific attribute of customers. On the basis of a specified threshold value for each key term the above matrix could be converted to binary matrix $BSKIM = \left[bs_j^d\right]$, where $bs_j^d = 1\ \tilde{s}_j^d \geq \beta_d$, and $bs_j^d = 0$ otherwise. β_d is the threshold value of the key term d. Hence, we can obtain the weight of each attribute using the following formula.

$$w_d^s = \frac{\sum_j bs_j^d}{\sum_j \sum_d bs_j^d} \quad \forall d = 1, \ldots, n_r \tag{8}$$

Then, the *S-Agent* designs a supplier-buyer incidence matrix as $SBIM^d = \tilde{Y}_{ij}'^d$, where $\tilde{Y}_{ij}'^d$ represents the preference fuzzy value of jth supplier for ith customer ($i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, k$) corresponding to the representative attribute d. To obtain the suppliers' profile matrix, we can calculate the average weight of $SBIM^d$ matrix as below:

$$MSBIM = \begin{bmatrix} \tilde{\beta}_{11} & \cdot & \cdot & \tilde{\beta}_{1k} \\ \tilde{\beta}_{21} & \cdot & \cdot & \tilde{\beta}_{2k} \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \tilde{\beta}_{m1} & \cdot & \cdot & \tilde{\beta}_{mk} \end{bmatrix} \tag{9}$$

where, $\tilde{\beta}_{ij} = \sum_{d=1}^n \tilde{Y}_{ij}'^d w_d^s$.

We observe that the value calculated for $\tilde{\beta}_{ij}$ as above represents the final priority level of jth supplier on ith customer.

Similar to *B-Agent*, the *S-Agent* requests the jth supplier to determine the set of objective functions, $CF_j^s = \{f_1^s, f_2^s, \ldots, f_l^s\}$, on behalf of corresponding firm.

The manner of building up profiles for buyers and suppliers in the network is graphically summarized in Fig. 3.

Coordinator Agent

During service integration, decision and coordination among services by agents are modeled as a distributed constraint satisfaction problem in which solutions and constraints are distributed into set of services and to be solved by a group agent. Finding a global solution to the composite service requires that all agents find the solutions that satisfy not only their own constraint but also inter-agent constraints. The interactive behavior of *B-Negotiator Agent* and *S-Negotiator Agent* involved in decision as well as the role of *Coordinator Agent* in coordination of e-SN is elaborated as follows.

Feasibility analyzer

Actual managerial decisions are made both based on preference priority and practical feasibility. With feasibility analyzer and considering the feasibility structure for transaction, we convert each buyers' and suppliers' profile into a decision matrix.

On the basis of a specified threshold value, we first convert the *MBSIM* and *MSBIM* matrices into binary matrices as $BMBSIM = \left[ma_{ij}\right]$, where $ma_{ij} = 1$ if $\tilde{a}_{ij} \geq \alpha_b$ and $ma_{ij} = 0$ otherwise, and $BMSBIM = \left[m\beta_{ij}\right]$, where $m\beta_{ij} = 1$ if $\tilde{\beta}_{ij} \geq \beta_s$ and $m\beta_{ij} = 0$. α_b and βs are the threshold values of buyers and suppliers viewpoint. Then, in order to determine the transactional counterparts in the network and also to obtain the interaction among buyers and suppliers, Supply Network Decision Matrix (*SNDM*) can be calculated by component-wise multiplication as a relative matching method (Mahdavi et al. 2009) as follows:
$SNDM = [dsn_{ij}]$,
where,

$$dsn_{ij} = ma_{ij} \times m\beta_{ij} \quad \begin{array}{l}\forall i = 1, \ldots, m \\ \forall j = 1, \ldots, k.\end{array} \tag{10}$$

The elements of the above matrix will classify the buyers' and suppliers' coalitions according to their points of view.

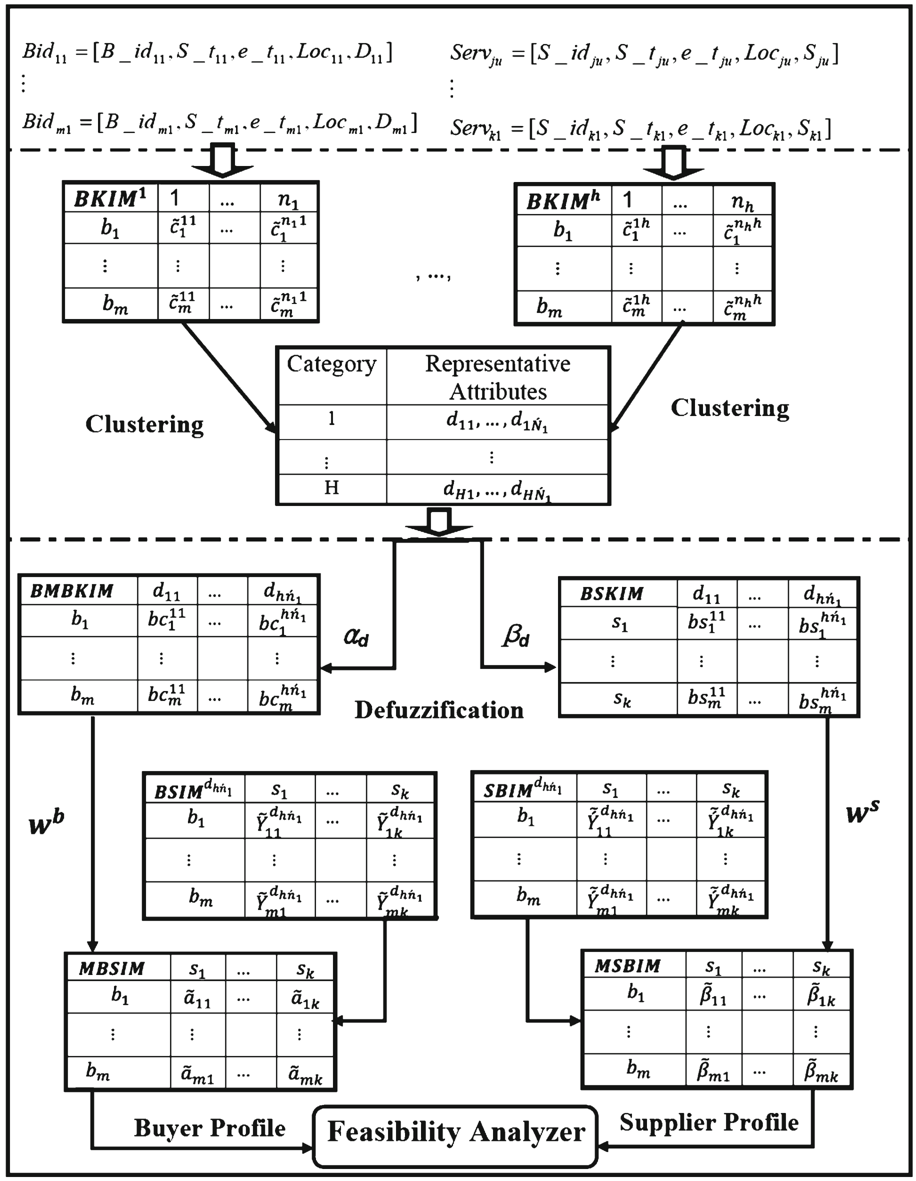

Fig. 3 Buyers and suppliers' profile in SN

Therefore, any two nodes of the network for which the element of *SNDM* is not zero, will participate in the negotiation process. In the next section, we discuss the network global model which is utilized to evaluate the goodness of proposals during the negotiation process.

Global solution

In a non-coordinated SN, the intention of each agent is to maximize its individual goal. On the other hand, in a coordinated SN, both buyers and suppliers would consider maximizing network profit as the locally optimal solutions do not guarantee a feasible transaction between partners. Therefore, the *Coordinator Agent,* after feasibility analysis, delivers a message to both buyers and suppliers and announces the feasible transactional counterparts. Then, by receiving the corresponding information including suggested reserved value, upper, and lower bounds for the price and lead time, the *Coordinator Agent* will refine the local constraints and variables to find a mutually compatible solution and evaluate the proposals during the negotiation process as well. In order to preserve social welfare, i.e. fulfilling all participants' objectives, in a multi-agent system, we define a mathematical model with global objective function which is the minimization of sum of the relative "errors" of all agents. The parameters and decision variable are presented as follows:

Parameters

L_{ij}^{q} : The lower bound of capacity which jth supplier can transfer to ith buyer for part family q.

e_{ij}^q : NPV (net present value) of unit transportation cost from jth supplier to ith buyer for part family q.
c_j^q : Inventory holding unit cost of jth supplier for part family q.
B_j : Maximum available operational budget (inventory holding and carrying costs) of jth supplier.
D_i^q : Demand of ith buyer for part family q.
S_j^q : Supply of jth supplier for part family q.
q: number of part families.
f_{lq}^{*b} : Optimal value of lth buyers' objective function for part family q before cooperation.
f_{lq}^{*s} : Optimal value of lth suppliers' objective function for part family q before cooperation.

Decision variables

X_{ij}^q : The decision variable giving the quantity of commodities which ith buyer buys from jth supplier for part family q (or equivalently, jth seller sells to ith buyer).

Multi-objective function

$Z = Min[P^b(X), P^s(X)]$,
where,

$$P^b(X) = \sum_q \sum_l \sum_i \sum_j \frac{f_{lq}^{*b} - f_{lq}^b\left(X_{ij}^q\right)}{f_{lq}^{*b}},$$

$$P^s(X) = \sum_q \sum_l \sum_i \sum_j \frac{f_{lq}^{*s} - f_{lq}^s\left(X_{ij}^q\right)}{f_{lq}^{*s}}. \tag{11}$$

Operational constraints

$$\begin{array}{lll}
\textbf{1}. & X_{ij}^q \ge L_{ij}^q & \forall i, j, q \\
\textbf{2}. & \sum_j X_{ij}^q = D_i^q & \forall i, q \\
\textbf{3}. & \sum_i X_{ij}^q \le S_j^q & \forall j, q \\
\textbf{4}. & \sum_q \left[\sum_i \left(e_{ij}^q \times X_{ij}^q\right)\right] & \\
 & +\sum_q \left[c_j^q \left(S_j^q - \sum_i X_{ij}^q\right)\right] \le Bj & \forall j \\
\textbf{5}. & X_{ij}^q \ge 0 & \forall i, j, q
\end{array} \tag{12}$$

The first constraint guarantees the system equilibrium so that each supplier meets the minimum requirement of each buyer. Constraints 2 and 3 are considered to satisfy the demand and supply. The forth one implies that operational costs including inventory holding and carrying costs should not exceed the maximum available budget for each supplier. The proposed model describes a multi-objective problem. Using weighting method to formulate a single objective function, non-dominated solutions can be obtained. This method guarantees the network profit through the selection of Pareto optimal solutions among agents. Cast in the classical constraint optimization literature, our problem is hence a distributed constraint optimization problem augmented with agent local objective functions. In fact, f_{lq}^{*b} and f_{lq}^{*s} can be obtained by solving a classical constrained optimization problem locally for each agent before coordination in the network. Note that the model introduced above is of a general form and some additional constraints may be added when dealing with some particular cases of practical relevance. In the next section, we discuss the process of proposed automated negotiation and illustrate how feasible partners negotiate until achieving mutual compatible solutions.

Dynamic automated negotiation model

In a real problem, buyers send RFQ messages and suppliers initially responds to the receipt of a buyer offer by evaluating the offer and considering their ability to satisfy delivery dates and prices. Hence, the automated negotiation is formulated from the perspective of the supplier. After the *Coordinator Agent* determined the material flow based on the Pareto optimal solutions, checks the buyers' and suppliers' reserved value for prices and lead times whether satisfy one another. If the suppliers' suggestions exceed the buyers' ones, the *Coordinator Agent* generate new offers to enhance the global welfare of the network.

The protocol

Negotiation protocols are defined as the set of rules which govern the interaction. Indeed, any negotiation is guided by a protocol, which describes the rules of the dispute, that is, how the parties exchange their offers, and how and when the negotiation can go on or terminate.

Information sets

B-Negotiator Agent and S-Negotiator Agent have private information set: $I^B = \langle P_R^B, P_{max}^B, LT_{max}^B \rangle$, $I^S = \langle P_R^S, P_{min}^S, LT_{min}^S \rangle$. P_R^B and P_R^S are reserved price of buyers and sellers, respectively. P_{max}^B and LT_{max}^B are the buyer's maximum price and lead time that he can offer, any proposal higher than it should not be accepted. For the seller, similarly, P_{min}^S and LT_{min}^S are the seller's minimum price and lead time.

Rule of bargain

First, *B-Negotiator Agent* sends a proposal to *Coordinator Agent.* Then, the agent evaluates the proposal through the global objective functions introduced in Sect. "Global solution". At time *t,* the *Coordinator Agent* generates new offers if the suppliers' suggestions exceed the buyers' ones. The agent action and choice at time $t+1$ follow the rule as shown below:

$$\text{Action}^{t+1} = \begin{cases} \text{Quit}, & \text{if } t \geq T_{max} \\ \text{Accept}, & \text{if } t < T_{max} \text{ and } Z_{t+1} \leq Z_t \\ \text{Generate new offers}, & \text{otherwise} \end{cases} \tag{13}$$

The proposal generation

This section describes how to generate an offer at a certain stage under the control of the protocol which described above. Review of the literature reveals that most studies have focused on the design of different negotiation models on the basis of the bargain on prices where the goodness of the generated offers is evaluated using utility function. The use of utility function would affect the outcomes of negotiations adversely if the appropriate one has not been applied. On the other hand, in practice, a supplier sells products to many buyers and a buyer may distribute its demand among many suppliers. Hence, the issue of quantity can complicate the decisions in a negotiation since it generally interacts with other issues such as price (Cheng et al. 2006) and lead time. Therefore, in this study, the price proposal is considered to be dependent on lead time and the goodness of the proposals is evaluated in terms of the material flow and total welfare of the network. Automobile and power generation industries are two main sectors which follow the above proposed objectives.

In fact, the suppliers' offered lead time is based on their capacity and operational constraints. In some cases, based on the buyers' requirements and the importance degree for the price attribute, the suppliers may be capable to deliver the products earlier with early delivery fee. In contrast, in order to decrease the workload in specific periods, some suppliers may offer a discount to the buyers such that agree to extend the lead time. Therefore, in practice the price proposal depends on the deviation of the buyers' and suppliers' desired lead times.

The rationale for generating the proposal is represented in Fig. 4:

Let P_{in}^{j} be the initial or reserved value of suggested price for jth supplier and $\Delta T = LT'_{ij} - LT_{ij}$ be the discrepancy of suggested lead times for jth supplier and ith buyer. ΔP is also the increment in price for each unit of ΔT.

In discrete states, the relation between price and lead time has usually been considered as a step function. However, in real market situations, the increment in price does not reveal linear behavior. Therefore, in continuous cases, exponential function can be the most suitable fit for the price function.

In this study, an exponential function has been used to define the price function (P) as follows:

$$P = \beta e^{\alpha \Delta T} \tag{14}$$

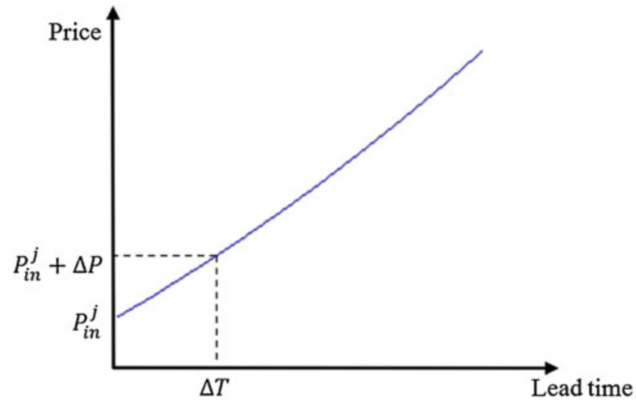

Fig. 4 Proposal function of the agent

By substituting the parameters and doing some mathematical calculations, the function can be obtained as follows:

$$P = P_{in}^{j}\left(\frac{P_{in}^{j} + \Delta P}{P_{in}^{j}}\right)^{\Delta T} \tag{15}$$

The concession degree for the attribute of price is also defined as shown below:

$$CD^{p} = \frac{\left(1 - CC_{j'}^{p}\right)}{\left(1 - CC_{j'}^{p}\right) + \left(1 - CC_{i'}^{p}\right)} \tag{16}$$

where, $CC_{j'}^{p}$ and $CC_{i'}^{p}$ are the defuzzified value of jth supplier and ith buyer' importance degree, respectively.

Consequently, the function for price proposal can be formulated as follows:

$$P(t) = P_{in}^{j}\left(\frac{P_{in}^{j} + \Delta P}{P_{in}^{j}}\right)^{\Delta T} - \left(P_{in}^{j}\left(\frac{P_{in}^{j} + \Delta P}{P_{in}^{j}}\right)^{\Delta T} - P_{min}^{j}\right)(t/T_{max})^{CD^{p}} \tag{17}$$

where, T_{max} is time deadline for the negotiation and P_{min}^{j} is the minimum value of the price that jth supplier can sell the products. In the above function, $(t/T_{max})^{CD^{p}}$ is the recession degree where the seller will give a small recession at the beginning and make bigger recession when t is approaching T_{max}. If $\Delta T > 0$, the function increases the initial price and in the case of $\Delta T < 0$, the function consider some discounts.

In order to propose the offers in each negotiation round that are likely to be accepted by the opponent, the negotiation rule can be constructed as:

$$NR = Rand\,[P(t+1), P(t)] \quad \forall t = 0, 1, \ldots, T_{max} \tag{18}$$

where, $Rand[a, b]$ is the random number between a and b.

Dynamic strategy

In order to get more profit, a 'wise' negotiator will generate a proposal that is decided not only by its information set, but also by the belief of the opponent's information set (Ke-xing et al. 2007). Therefore, during the process of bargain, the agent changes its belief dynamically, until reach an agreement of terminate. The algorithm of proposed negotiation process is represented in Fig. 5.

The negotiation among interacting counterparts would be withdrawn if the offers are infeasible for a certain number of successive negotiation rounds, where this number is specified by the user. In the next section, we first apply the proposed model in a network scenario and then evaluate its performance using simulation model.

Validation and performance evaluation

Consider a scenario in which the sellers sell a product. Every day, customers send RFQ and suppliers bid on them depending on their ability to satisfy the customers' delivery dates and prices. The demand will be fulfilled if there is available and sufficient capacity. Otherwise, it will be supplied in the course of backlog orders.

To examine the feasibility of the model, a prototype system was implemented using *Matlab* software. The starting assumption is that first suppliers register in the network by informing their capacities and other attributes such as location, product scope and their specifications and so on. Thereby, their information will be set before simulation. The model begins by generating buyers randomly. In addition, it is assumed that each supplier has designed four initial categories of attributes which include "Delivery and Customer Service", "Price", "Quality" and "Manufacturing Domain" as described in Table 1. Usually a supply link has to be embedded into the entire SN. For instance, the supplier might not have just one but several other producers he/she is in contact with. In capturing this influence of the full network, the availability of the suppliers' capacity is considered to be stochastic. We also consider the production planning as "make to order". The total cost of SN entities is composed of suppliers' unutilized capacity and backlog orders.

In the remaining part of this section, we conduct experiments to validate the proposed coordination mechanism for the SNs. We compare the performance of our model with that of nowadays tendering process which is being practiced by industries. The comparison is conducted in terms of total network cost. The suggested approach relies on an agent-based model and follows the coordination mechanism while the other method does not employ the coordination mechanism. In order to investigate the performance of the proposed approach, the following hypotheses are considered:

H_0: The proposed intelligent mechanism outperforms the normal tendering process in terms of backlog order costs.
H_0: The proposed intelligent mechanism outperforms the normal tendering process in terms of unutilized capacity costs.

The statistical analyses of the above addressed hypotheses are performed through one-way ANOVA.

Experiment: performance evaluation

In this study, two scenarios have been investigated. In the first scenario, the proposed approach is considered. This includes a dynamic decision making system whereby the attribute evaluation are generated based on the suppliers' past performance for satisfying buyers' requirements. At each stage, the amount of supplies and corresponding offered lead times of each supplier are set based on the available capacity and the utilized capacity in the previous negotiation rounds.

The second scenario, follows a normal tendering process where the buyers announce the tender by giving the scope and scale of the tender along with their desired due dates. Then, the suppliers bid on them by giving their proposed price and delivery times so that they can fulfill the other attribute values contained in the tender document. In this scenario, the parties are not allowed to share information and hence the partners are selected based on the similarity in the proposed price and lead time attributes. In each round, the suppliers reveal their reserved value, upper and lower bounds for the price and lead time which are dependent on the capacity and operational constraints. It is also assumed that if a supplier had backlog orders for two sequential periods, a defensive strategy will be applied by the suppliers through feasible reduction of the price to maintain the market share.

The performance of the both scenarios is measured using network cost in terms of the cost associated with backlog orders (Fig. 5), suppliers' unutilized capacity (Fig. 6), and the total cost which is the aggregation of the aforementioned costs (Fig. 7). The unit cost of SN entities is shown in Table 3. It should be noted that the SN cost is aggregated through all the players. Each decision level is simulated 100 times with a 40 period time span for each simulation in order to obtain a steady state results. In each period, a uniform distribution in a range between 1 and 20 is used to generate the number of the buyers. The supply, customer's demand, suppliers' suggested lead times and prices are also generated using a uniform distribution. The buyers' suggested lead times and prices follow a normal distribution. With a predefined probability, the previous buyers enter the system in some rounds of simulation. It is also assumed that, in some cases, suppliers are not capable to deliver the products on time. Hence, the demand is supplied in the course of backlog order.

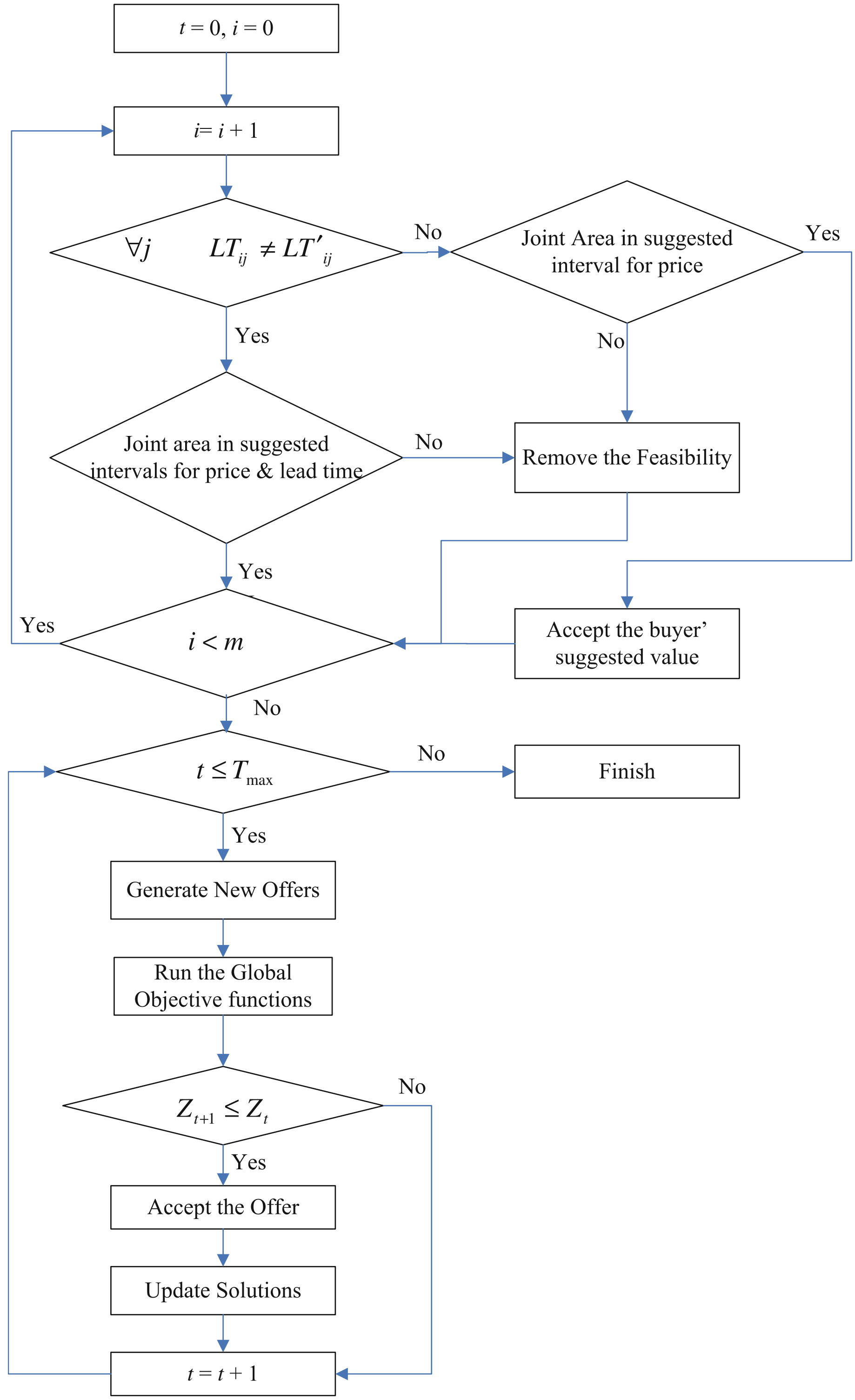

Fig. 5 The proposed negotiation process algorithm

The parameters used in the simulation model are set as follows:

- $T_{max} = 10$
- If $\Delta T > 0 \rightarrow \Delta P = 5\%$, If $\Delta T < 0 \rightarrow \Delta P = 2\%$
- $w^b = w^s = 0.5$
- $\alpha_d = \beta_d = 0.4$

Discussion and analysis

Figures 6, 7, and 8 demonstrate the performance of the proposed approach and that of the normal tendering process in terms of backlog cost, unutilized capacity cost, and total network cost, respectively. The performance measure

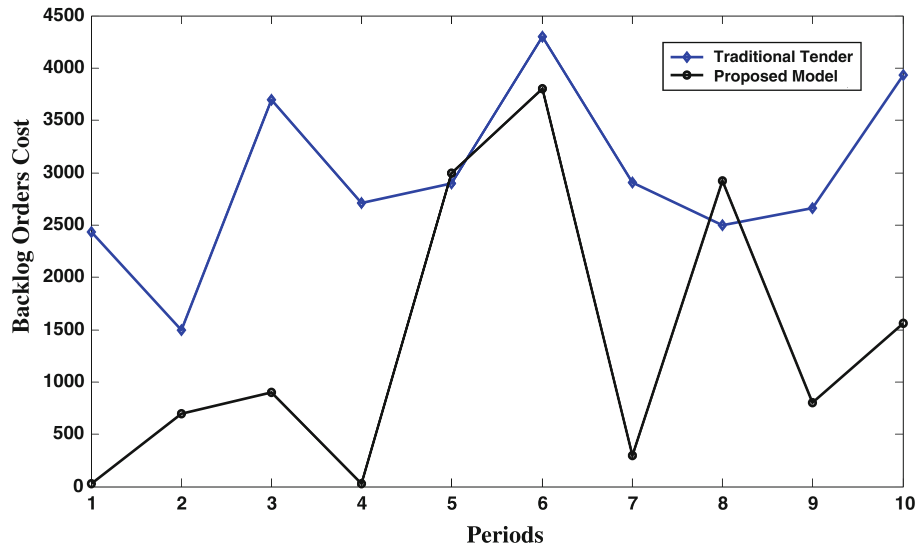

Fig. 6 Backlog order cost curve of proposed model versus traditional tendering process

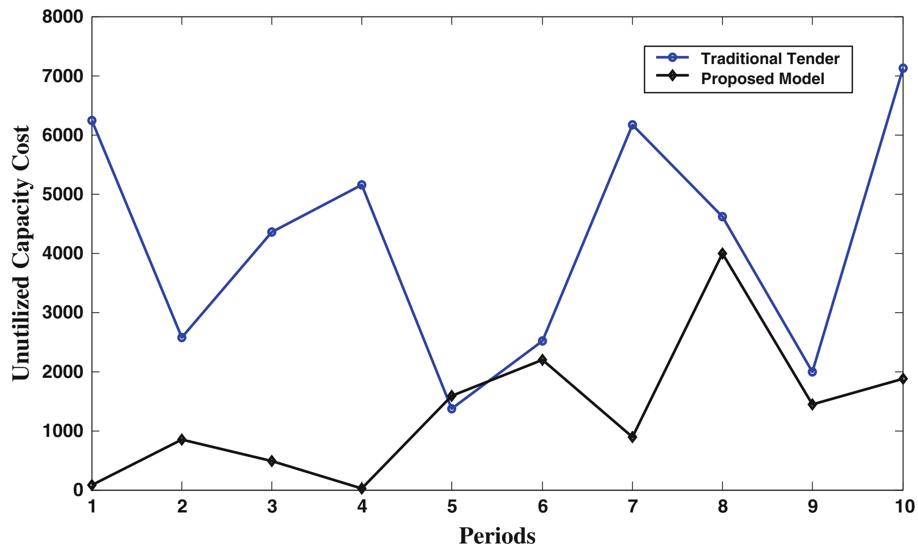

Fig. 7 Unutilized capacity cost curve of proposed model versus traditional tendering process

Table 3 Unit costs

Unutilized capacity unit cost per period	Backlog order unit cost per period
3	5

associated with backlog order cost represents the buyers' satisfaction while the unutilized capacity indicates the supplier shop floor utilization. In general, the results indicate that in all three cases, the proposed approach outperforms the normal tendering process. This is mainly due to the intelligent behavior of the proposed approach where both coordination and negotiations among the buyers and suppliers have been taken into account. In addition, this has made it possible to utilize the suppliers' capacity more efficiently while minimizing the backlog orders. From the scholars view point this is one of the main challenge which needs more focus in enhancing the performance of a SN.

In order to investigate the significance of the presented hypotheses in Sect. "Validation and performance evaluation", the one-way ANOVA is applied. The results as presented in Table 4, supports both hypotheses at $p \leq 0.01$. Since the null hypothesis in ANOVA is rejected at the significance level of 0.05, it is desired to have a pair-wise comparison of factor level means in order to investigate the factor effects. Considering the obtained confidence intervals for differences between the factor level means $\left(\left(\mu_{Agent} - \mu_{Tender}\right)\right)$, normal tendering process leads to higher mean costs in the network. These outcomes approve that the proposed approach leads to both reliable and higher performance than that of the normal tendering approach.

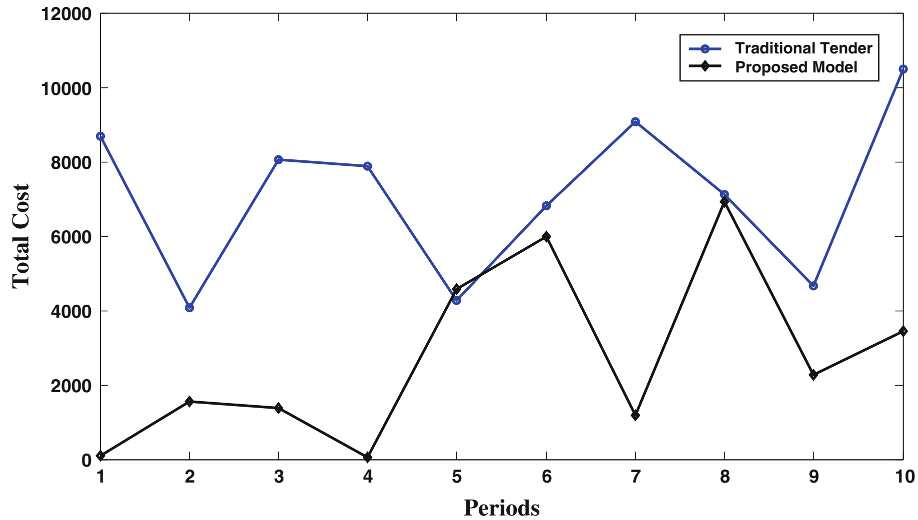

Fig. 8 Total cost curve of proposed model versus traditional tendering process

Table 4 One-way anova table

Hypothesis	Performance	F-statistics	P-value	Confidence interval
H_0 (agent > tender)	Backlog order cost	8.03	0.01	$(-2.59 \times 10^3, -0.38 \times 10^3)$
H_0 (agent > tender)	Unutilized capacity cost	15.18	0.0011	$(-4.41 \times 10^3, -1.32 \times 10^3)$

Conclusions

The world of business is being changed to an e-economy by new forces driven from global competition, increased information availability, changing relationships, rapid innovations, and increasingly complex products. Hence, traditional linear supply chain systems are converted into network systems in which collaboration has been significantly improved. In addition, the advent of internet based market places motivated researchers to adopt conceptual buyer/supplier coordination models to save the system costs and ultimately improve the performance of the SN.

In this study, we developed a framework based on multi-agent systems to facilitate collaboration and negotiation in dynamic environments. To establish this mechanism, we provided a novel approach in designing a rich profile for buyers and suppliers. The next contribution was the design of feasibility analyzer module to cluster transactional counterparts while considering each entity profile. The use of utility function for determining the most promising partners would affect the outcomes of negotiations adversely if the appropriate one has not been applied. In addition, the proposed negotiation model differs from others that have been presented. First, it promises the agents change the negotiation strategies dynamically and takes the time constraints of bargainers into account. This is more flexible especially in e-based SNs. Second, the goodness of the generated offers is evaluated in terms of the material flow and the welfare of whole network. The proposed approach present potentials to resolve conflicting problems in real-world SNs. Accordingly, the validation results approved the importance of the proposed coordination mechanism in SNs.

Further research may consider more uncertainty in the network. Random number of suppliers and expansion of the network by including more echelons like manufacturers and retailers to extract the optimal strategies are interesting to consider for coordination improvement in a SN. In addition, deep concentration on dynamic aspects of the network to evaluate the impact of strategies on competitive advantages is a new perspective that can be lead to more practical outcomes.

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Appendix

The determination of the number of clusters is the most important issue in clustering. There are many studies on this issue. Here, the following criterion is used for this purpose (Sugeno and Yasukawa 1993):

$$S^{r}(o)=\sum_{i=1}^{m}\sum_{f=1}^{o}\left(\mu_{fi}^{r}\right)^{m'}\left(\left\|c_{i}^{dr}-v_{f}^{r}\right\|^{2}-\left\|v_{f}^{r}-\bar{c}^{r}\right\|^{2}\right) \tag{19}$$

where

m: number of data to be clustered;
o: number of clusters, $c \geq 2$;
$\bar{c}^{r}$: average of data: $c_1^{dr}, c_2^{dr}, \ldots, c_m^{dr}$;
v_f^r: vector expressing the center of the fth cluster;
$\|\circ\|$: norm;
m': adjustable weight (usually $m = 1.5 \sim 3$)

The number of clusters, o, is determined so that $S^r(o)$ reaches a minimum as o increases: it is supposed to be a local minimum as usual. The first term of the right-hand side is the variance of the data in a cluster and the second term is that of the clusters themselves. Therefore the optimal clustering is considered to minimize the variance in each cluster and to maximize the variance between the clusters.

By defining a threshold value, ξ, the members of each cluster can be identified from matrix U.

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